



ZYNQ™ ULTRASCALE+™ PRODUCT SELECTION GUIDE

PS	Device Name		ZU21DR	ZU25DR	ZU27DR	ZU28DR	ZU29DR	ZU39DR	ZU42DR		ZU43DR	ZU46DR		ZU47DR	ZU48DR	ZU49DR	
	Gen 1							Gen 2		Gen 3							
	Quad-core Arm® Cortex®-A53 MPCore™ up to 1.3GHz, Dual-core Arm Cortex-R5F MPCore up to 533MHz																
RF Data Converter	12-bit RF-ADC w/DDC	# of ADCs	0	8	8	8	16	16	–		–	–		–	–	–	
		Max Rate (GSPS)	0	4.096	4.096	4.096	2.058	2.220	–		–	–		–	–	–	
	14-bit RF-ADC w/DDC	# of ADCs	–	–	–	–	–	–	8	2	4	8	4	8	8	16	
		Max Rate (GSPS)	–	–	–	–	–	–	2.5	5.0	5.0	2.5	5.0	5.0	5.0	2.5	
	14-bit RF-DAC w/DUC	# of DACs	0	8	8	8	16	16	8	4		12		8	8	16	
		Max Rate (GSPS)	0	6.554	6.554	6.554	6.554	6.554	9.85 ⁽³⁾		9.85 ⁽³⁾	9.85 ⁽³⁾		9.85 ⁽³⁾	9.85 ⁽³⁾	9.85 ⁽³⁾	
	SD-FEC		8	0	0	8	0	0	0		0	8		0	8	0	
	Digital Front-End (DFE)		–	–	–	–	–	–	–		–	–		–	–	–	
	Number of DDCs per RF-ADC ⁽¹⁾		0	1	1	1	1	1	1		2	1		1	1	1	
	RF input Freq max. GHz		4					5	6								
	Decimation / Interpolation		1x, 2x, 4x, 8x					1x, 2x, 4x, 8x	1x, 2x, 3x, 4x, 5x, 6x, 8x, 10x, 12x, 16x, 20x, 24x, 40x								
	System Logic Cells (K)		930	678	930	930	930	930	489		930	930		930	930	930	
	CLB LUTs (K)		425	310	425	425	425	425	224		425	425		425	425	425	
	Max. Dist. RAM (Mb)		13.0	9.6	13.0	13.0	13.0	13.0	6.8		13.0	13.0		13.0	13.0	13.0	
	Total Block RAM (Mb)		38.0	27.8	38.0	38.0	38.0	38.0	22.8		38.0	38.0		38.0	38.0	38.0	
	UltraRAM (Mb)		22.5	13.5	22.5	22.5	22.5	22.5	45.0		22.5	22.5		22.5	22.5	22.5	
	DSP Slices		4,272	3,145	4,272	4,272	4,272	4,272	1,872		4,272	4,272		4,272	4,272	4,272	
	GTy Transceivers		16	8	16	16	16	16	8		16	16		16	16	16	
	PCIe® Gen3 x16		2	1	2	2	2	2	–		–	–		–	–	–	
	PCIe® Gen3 x16/Gen4 x8 / CCIX ⁽²⁾		–	–	–	–	–	–	0		2	2		2	2	2	
	150G Interlaken		1	1	1	1	1	1	0		1	1		1	1	1	
	100G Ethernet MAC/PCS w/RS-FEC		2	1	2	2	2	2	0		2	2		2	2	2	
	System Monitor		2	2	2	2	2	2	2		2	2		2	2	2	
	Speed Grades		-1E, -1I, -1LI, -2E, -2LE, -2I, -2LI	-1E, -1I, -1LI, -2E, -2LE, -2I, -2LI	-1E, -1I, -1LI, -2E, -2LE, -2I, -2LI	-1E, -1I, -1LI, -2E, -2LE, -2I, -2LI	-1E, -1I, -1LI, -2E, -2LE, -2I, -2LI	-1E, -1I, -1LI, -2E, -2LE, -2I, -2LI	-2I, -2LI		-1E, -1I, -1LI, -2E, -2I, -2LI	-1E, -1I, -1LI, -2E, -2I, -2LI	-1E, -1I, -1LI, -2E, -2I, -2LI		-1E, -1I, -1LI, -2E, -2I, -2LI	-1E, -1I, -1LI, -2E, -2I, -2LI	
Programmable Logic (PL)	Package Footprint	Package Dimensions	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC		
	D1156	35x35	214, 72, 208 4, 16 0, 0														
	E1156	35x35		214, 48, 104 4, 8 8, 8	214, 48, 104 4, 8 8, 8	214, 48, 104 4, 8 8, 8			214, 24, 128 4, 8 10, 8	214, 48, 104 4, 8 4, 4		214, 48, 104 4, 8 8, 8	214, 48, 104 4, 8 8, 8				
	G1517	40x40		214, 48, 299 4, 8 8, 8	214, 48, 299 4, 16 8, 8	214, 48, 299 4, 16 8, 8				214, 48, 299 4, 16 4, 4		214, 48, 299 4, 16 8, 8	214, 48, 299 4, 16 8, 8				
	F1760	42.5x42.5		214, 96, 312 4, 16 16, 16			214, 96, 312 4, 16 16, 16										214, 96, 312 4, 16 16, 16
	H1760	42.5x42.5									214, 48, 312 4, 16 12, 12						

zylin™ UltraScale™ RF SoCs

1. This value applies when all RF I/O of an RF-ADC tile are used.

2. Operates in compatibility mode for 16.0GT/s (Gen4) operation. See [PG213](#).

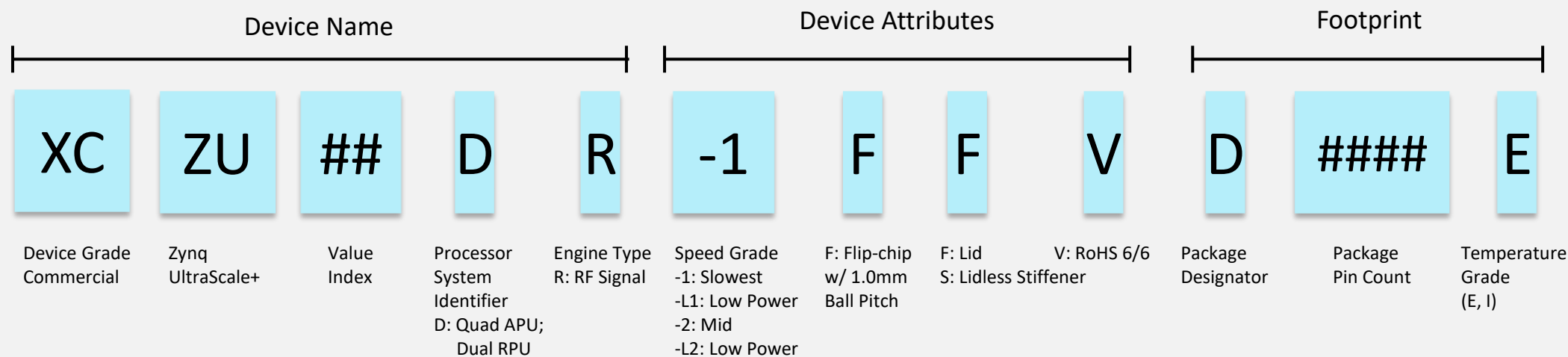
3. For operation up to 10GSPS, contact your local Xilinx Sales Representative.

XMP105 (v1.12.1)

		Device Name	ZU63DR	ZU64DR	ZU65DR	ZU67DR			
			DFE						
PS			Quad-core Arm® Cortex®-A53 MPCore™ up to 1.3GHz, Dual-core Arm Cortex-R5F MPCore up to 533MHz						
	14-bit RF-ADC w/DDC	# of ADCs	4	2	8	2	6	8	2
RF Data Converter		Max Rate (GSPS)	2.95	5.9	2.95	5.9	5.9	2.95	5.9
	14-bit RF-DAC w/DUC	# of DACs	4		8		6		8
Programmable Logic (PL)		Max Rate (GSPS)	10.0 ⁽³⁾		10.0 ⁽³⁾		10.0 ⁽³⁾		10.0 ⁽³⁾
		SD-FEC	0		0		0		0
		Digital Front-End Hard IP (DFE IP)	Channel Filter, DUC/DDC, Mixer, CFR, Complex Equalizer, PQ Resampler, DPD						
		Low PHY Hard IP	FFT/iFFT, PRACH		None		FFT/iFFT, PRACH		FFT/iFFT, PRACH
		Number of DDCs per RF-ADC ⁽¹⁾	1		1		1		1
		RF input Freq max. GHz	7.125						
		Decimation / Interpolation	1x, 2x, 3x, 4x, 5x, 6x, 8x, 10x, 12x, 16x, 20x, 24x, 40x						
		System Logic Cells (K)	393		328		489		489
		CLB LUTs (K)	180		150		224		224
		Max. Dist. RAM (Mb)	5.47		4.56		6.8		6.8
		Total Block RAM (Mb)	17.6		15.8		22.8		22.8
		UltraRAM (Mb)	36.6		22.5		45.0		45.0
		DSP Slices	1,200		1,872		1,872		1,872
		GTY Transceivers	4		8		8		8
		PCIe® Gen3 x16/Gen4 x8 / CCIX ⁽²⁾	0		0		0		0
		150G Interlaken	0		0		0		0
		100G Ethernet MAC/PCS w/RS-FEC	1		1		1		1
		System Monitor	2		2		2		2
		Speed Grades	-1I, -1LI, -2I, -2LI		-1I, -1LI, -2I, -2LI		-1I, -1LI, -2I, -2LI		-1I, -1LI, -2I, -2LI
	Package Footprint	Package Dimensions	PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC		PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC		PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC		PSIO, HDIO, HPIO GTR, GTY RF-ADC, RF-DAC
	E1156	35x35	214, 24, 130 4, 4 6, 4		214, 24, 130 4, 8 10, 8		214, 24, 130 4, 8 6, 6		214, 24, 130 4, 8 10, 8

Notes:
1. This value applies when all RF I/O of an RF-ADC tile are used.
2. Operates in compatibility mode for 16.0GT/s (Gen4) operation. See [PG213](#).
3. 10GSPS RF-DAC operation is available in -2I speed grade.

Zynq® UltraScale+™ RFSoc Ordering Information



E = Extended ($T_j = 0^{\circ}\text{C}$ to $+100^{\circ}\text{C}$)
 I = Industrial ($T_j = -40^{\circ}\text{C}$ to $+100^{\circ}\text{C}$)

Note: -L2E ($T_j = 0^{\circ}\text{C}$ to $+110^{\circ}\text{C}$); -L2I ($T_j = -40^{\circ}\text{C}$ to $+110^{\circ}\text{C}$)
 Refer to [DS889](#), Zynq UltraScale+ RFSoc Data Sheet: Overview for additional information

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